

# KNIGHT.VAULT — USER MANUAL & RUNTIME SPECIFICATION

A Comprehensive Handbook for Operating, Deploying, and Troubleshooting Confidential Zero-Knowledge Settlement Vaults on Midnight Network

|                                                           |                                                           |
|-----------------------------------------------------------|-----------------------------------------------------------|
| <b>Target Network:</b> Midnight Preprod                   | <b>DApp Name:</b> KNIGHT.VAULT                            |
| <b>Smart Contract:</b> Compact Circuits (payment.compact) | <b>ZK Proof Engine:</b> Docker Local Port :6300 / Railway |
| <b>Wallet API:</b> Midnight Lace DApp Connector           | <b>Architect:</b> Kshitij Adakane                         |

## 1. Developer Setup & Prerequisites

To execute transactions and run proof synthesis locally, the developer environment requires three primary subsystems:

- **WSL2 (Ubuntu 22.04 LTS):** Required on Windows to host the native Linux Compact toolchain and prover libraries.
- **Docker ProofServer (Port 6300):** Synthesizes zero-knowledge proofs on your machine in 2-4 seconds.
- **Midnight Lace Browser Wallet:** Manages unshielded/shielded keys, signs transactions, and relays sponsored dust fees.

### 1.1 Docker ProofServer Deployment & Port Verification

Run the proof server container and ensure it is listening on localhost:

```
docker run -d --name proof-server -p 6300:6300 midnightnetwork/proof-server:latest
# Health check command:
curl http://localhost:6300/health
```

**WSL2 Memory Note:** Ensure your .wslconfig has at least 8GB-12GB RAM allocated. Proving large ZK circuits may fail with an unhandled exit code if memory is constrained below 4GB.

## 2. User Operations: Deposit & Withdraw

### 2.1 Depositing Funds (Zero-Gas Payer Flow)

1. Open the KNIGHT.VAULT dashboard at <http://localhost:5173/>.
2. Click '**Connect Wallet**' and authorize permissions on the popup modal.
3. Enter the deposit amount (e.g. 50 tNIGHT) in the '**Inflow Channel**' card.
4. Click '**Execute Confidential Inflow**'. The Lace wallet balances the transaction with sponsored dust automatically.

### 2.2 Zero-Knowledge Withdrawal Flow

1. Navigate to the '**Private Liquidity Release**' card.

2. Enter the withdrawal amount and recipient unshielded address (or public key).
3. Click '**Authorize ZK Withdrawal**'. The local ProofServer will compute the witness proof.
4. Once synthesized, the transaction is committed to Midnight Preprod with zero private data leaked.

### 3. Real-World Troubleshooting & FAQs

| Issue / Error                              | Root Cause & Recommended Solution                                                                                                                                    |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ProofServer Unreachable (Port 6300)</b> | Docker container is stopped or port is blocked by Windows Firewall. Run docker restart proof-server and verify with curl <code>http://localhost:6300/health</code> . |
| <b>Lace Wallet 'offset: null' Error</b>    | Known bug in Midnight Indexer v4 GraphQL schema when pagination offset is null. The KNIGHT.VAULT client includes an automatic filter polyfill.                       |
| <b>Insufficient Dust Balance</b>           | The Lace wallet needs a small unshielded balance to derive dust. Request 10 tNIGHT from the Midnight faucet to refresh coin notes.                                   |
| <b>Vite / Browser Polyfill Errors</b>      | Node.js stream/crypto modules must be polyfilled in browser builds. Ensure vite-plugin-node-polyfills is included in vite.config.ts.                                 |

### 4. Summary & Architect Credits

KNIGHT.VAULT demonstrates modern, institutional-grade zero-knowledge engineering. Designed and built by **Kshitij Adakane**.

**Open Source Codebase:** <https://github.com/KSHITIJAdakane/KNIGHT.VAULT>